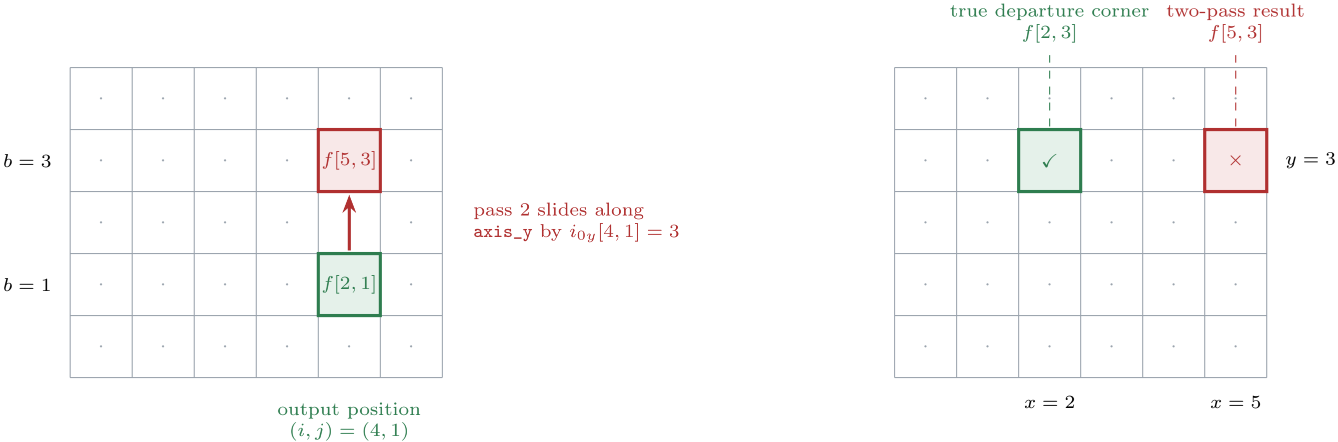
(a) after pass 1 along `axis_x`:
$t[a, b] = f\left[i_{0x}[a, b],\ b\right]$

(b) the array f, indexed $[x, y]$



Output position $(i, j) = (4, 1)$, with $i_{0x}[4, 1] = 2$, $i_{0y}[4, 1] = 3$, while $i_{0x}[4, 3] = 5$ is a different entry of the same index field.

two passes: $$\mathrm{out}[i, j] \;=\; t\left[i,\ i_{0y}[i, j]\right] \;=\; f\Big[\underbrace{i_{0x}\left[i,\ i_{0y}[i, j]\right]}_{\text{evaluated at the shifted position}},\ i_{0y}[i, j]\Big] \;=\; f[5, 3]$$

wanted: $$\mathrm{out}[i, j] \;=\; f\left[i_{0x}[i, j],\ i_{0y}[i, j]\right] \;=\; f[2, 3]$$